Exercises 5.5 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 33, 35, 39, 41, 45, 47, 57, 59, 61, 65, 67, 69, 71, 73, 77, 81, 87, 91, 93.

7. $\int \frac{\cos\sqrt{t}}{\sqrt{t}}\,dt, \quad u = \sqrt{t}$

$\frac{du}{dt} = \frac{1}{2\sqrt{t}}$; $dt = 2\sqrt{t}\,du$

$\int \frac{\cos u}{\sqrt{t}} \cdot 2\sqrt{t}\,du$

$= 2\int \cos u\,du$

$= 2\sin u + C = 2\sin\sqrt{t} + C$

9. $\int x\sqrt{1-x^2}\,dx$

Let $u = 1-x^2$; $\frac{du}{dx} = -2x$

$dx = \frac{1}{-2x}\,du$

$\int x\sqrt{u} \cdot \frac{1}{-2x}\,du$

$= -\frac{1}{2}\int \sqrt{u}\,du$

$= -\frac{1}{2} \cdot \frac{2}{3} \cdot u^{\frac{3}{2}} + C$

$= -\frac{1}{3}(1-x^2)^{\frac{3}{2}} + C$

11. $\int t^3 e^{-t^4}\,dt$

$= -\frac{1}{4}\int e^{-t^4}\,d(-t^4)$

$= -\frac{1}{4}e^{-t^4} + C$

13. $\int \sin(\pi t/3)\,dt$

Let $u = \frac{\pi t}{3}$, $\frac{du}{dt} = \frac{\pi}{3}$ → $dt = \frac{3}{\pi}\,du$

$= \int \sin u \cdot \frac{3}{\pi}\,du$

$= -\frac{3}{\pi}\cos u + C$

$= -\frac{3}{\pi}\cos\left(\frac{\pi t}{3}\right) + C$

15. $\int \frac{dx}{4x+7}$

let $u = 4x+7$, $\frac{du}{dx} = 4$, $dx = \frac{1}{4}du$

$\int \frac{1}{u} \cdot \frac{1}{4}\, du$

$= \frac{1}{4}\ln|u| + C$

$= \frac{1}{4}\ln|4x+7| + C$

17. $\int \frac{\cos\theta}{1+\sin\theta}\, d\theta$

let $u = 1+\sin\theta$, $\frac{du}{d\theta} = +\cos\theta$

$d\theta = +\frac{du}{\cos\theta}$

$\int \frac{\cos\theta}{u} \cdot \left(+\frac{1}{\cos\theta}\right) du$

$= +\int \frac{1}{u}\, du = +\ln|u| + C$

$= +\ln|1+\sin\theta| + C$

19. $\int \cos^3\theta \sin\theta\, d\theta$

let $u = \cos\theta$, $\frac{du}{d\theta} = -\sin\theta$

$d\theta = -\frac{du}{\sin\theta}$

$\int u^3 \cdot \sin\theta \cdot \left(-\frac{1}{\sin\theta}\right) du$

$= -\int u^3\, du = -\frac{u^4}{4} + C$

$= -\frac{\cos^4\theta}{4} + C$

21. $\int \frac{e^u}{(1-e^u)^2}\, du$

let $x = 1-e^u$, $\frac{dx}{du} = -e^u \Rightarrow du = -\frac{dx}{e^u}$

$= -\int \frac{e^u}{x^2} \cdot \left(\frac{1}{e^u}\right) dx$

$= -\int \frac{1}{x^2}\, dx = +x^{-1} + C = \frac{1}{1-e^u} + C$

23. $\int \frac{a + bx^2}{\sqrt{3ax + bx^3}}\,dx$

Let $u = 3ax + bx^3$.

$\frac{du}{dx} = 3a + 3bx^2$

$dx = \frac{du}{3a+3bx^2}$

$= \int \frac{a+bx^2}{\sqrt{u}} \cdot \frac{du}{3(a+bx^2)}$

$= \frac{1}{3}\int \frac{1}{\sqrt{u}}\,du$

$= \frac{2}{3} u^{\frac{1}{2}} + C$ $= \frac{2}{3}(3ax+bx^3)^{\frac{1}{2}} + C$

25. $\int \frac{(\ln x)^2}{x}\,dx$

Let $u = \ln x$; $\frac{du}{dx} = \frac{1}{x}$, $dx = x \cdot du$

$= \int \frac{u^2}{x} \cdot x\,du$

$= \int u^2\,du = \frac{u^3}{3} + C$

$= \frac{(\ln x)^3}{3} + C$

27. $\int \sec^2\theta \tan^3\theta\,d\theta$

Let $u = \sec\theta$; $\frac{du}{d\theta} = \sec\theta\tan\theta$

$d\theta = \frac{du}{\sec\theta\tan\theta}$

$= \int u \cdot \sec\theta\tan$

Let $u = \tan\theta$; $\frac{du}{d\theta} = \sec^2\theta \Rightarrow d\theta = \frac{du}{\sec^2\theta}$

$= \int \sec^2\theta \cdot u^3 \cdot \frac{du}{\sec^2\theta}$

$= \int u^3\,du = \frac{u^4}{4} + C = \frac{\tan^4\theta}{4} + C$

$\tan x$	$\sec^2 x$
$\sec x$	$\sec\tan x$
$\csc x$	$-\csc x \cot x$
$\cot x$	$-\csc^2 x$

tanx	sec²x
secx	secxtanx
cscx	−cscxcotx
cotx	−csc²x

33. $\int \frac{\sec^2\theta}{\tan\theta}\, d\theta$

$-\tan x$	$\sec^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$
$\cot x$	$-\csc^2 x$

let $u = \tan\theta$; $\frac{du}{d\theta} = \sec^2\theta$; $d\theta = \frac{du}{\sec^2\theta}$

$= \int \frac{\sec^2\theta}{u} \cdot \frac{du}{\sec^2\theta}$

$= \int \frac{1}{u}\, du$

$= \ln|u| + C = \ln|\tan\theta| + C$

35. $\int \frac{(\arctan x)^2}{x^2+1}\, dx$

- $D_x \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$ (inverse of sin)
- $D_x \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$
- $D_x \tan^{-1} x = \frac{1}{1+x^2}$

let $u = \tan^{-1} x$; $\frac{du}{dx} = \frac{1}{1+x^2}$

$dx = 1+x^2\, du$

$\int \frac{u^2}{x^2+1} \cdot x^2+1\, du$

$= \int u^2\, du = \frac{u^3}{3} + C = \frac{(\tan^{-1} x)^3}{3} + C$

39. $\int \cos(1 + 5t)\, dt$

let $u = 1+5t$; $\frac{du}{dt} = 5$; $dt = \frac{du}{5}$

$= \int \cos u \cdot \frac{1}{5}\, du$

$= \frac{1}{5} \int \cos u\, du = \frac{1}{5} \sin(1+5t) + C$

41. $\int \sqrt{\cot x}\, \csc^2 x\, dx$

let $u = \cot x$; $\frac{du}{dx} = -\csc^2 x$;

$dx = \frac{du}{-\csc^2 x}$

$= \int \sqrt{u} \cdot \csc^2 x \cdot \frac{du}{-\csc^2 x}$

$= -\int \sqrt{u}\, du = -\frac{2}{3} u^{\frac{3}{2}} + C$

$= -\frac{2}{3} \cot x^{\frac{3}{2}} + C$

$\tan x$	$\sec^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$
$\cot x$	$-\csc^2 x$

$u^{\frac{1}{2}}$ $\quad$ $\frac{2}{3} u^{\frac{3}{2}}$

45. $\int \frac{\sin 2x}{1+\cos^2 x}\,dx$

$\cos 2A = \cos^2 A - \sin^2 A$

$\cos 2A = 2\cos^2 A - 1$

$\cos^2 A = \frac{\cos 2A + 1}{2}$

先把 2次 ↓ 1次

$= \int \frac{\sin 2x}{\frac{3}{2} + \frac{1}{2}\cos 2x}\,dx$

Let $u = \cos 2x$; $\frac{du}{dx} = -2\sin 2x$; $dx = \frac{du}{-2\sin 2x}$

$= \int \frac{\sin 2x}{\frac{3}{2} + \frac{1}{2}u} \cdot \frac{du}{-2\sin 2x}$

$= -\frac{1}{2}\int \frac{1}{\frac{3+u}{2}}\,du$

$= -\frac{1}{2}\times 2\int \frac{1}{3+u}\,du$

$= -(3u + \frac{u^2}{2}) + C$

$= -3\cos 2x - \frac{\cos^2 2x}{2} + C$

$-\int \frac{1}{3+u}\,du = -\ln|3+u| + C$

$= -\ln|3+\cos 2x| + C$

47. $\int \cot x \, dx$

$\tan x \quad \sec^2 x$

$\sec x \quad \sec x \tan x$

$\cot x \quad -\csc^2 x$

$\csc x \quad -\csc x \cot x$

$$\int \frac{\cos x}{\sin x} \, dx$$

let $u = \sin x$; $dx = \frac{du}{\cos x}$

$$\int \frac{\cos x}{u} \cdot \frac{du}{\cos x}$$

$$= \int \frac{1}{u} \, du$$

$$= \ln|u| + C = \ln|\sin x| + C$$

57. $\int e^{\cos x} \sin x \, dx$

Let $u = e^{\cos x}$; $\frac{du}{dx} = -e^{\cos x} \cdot \sin x$; $dx = \frac{du}{-e^{\cos x} \cdot \sin x}$

$$\int u \sin x \cdot \frac{du}{-u \sin x}$$

$$= -\int du$$

$$= -u + c = -e^{\cos x} + c$$

59. $\int_0^1 \cos(\pi t/2)\, dt$

let $u = \frac{\pi}{2} t$; $\frac{du}{dt} = \frac{\pi}{2}$; $dt = \frac{2}{\pi} \cdot du$

when $t = 1$, $u = \frac{\pi}{2}$

$t = 0$, $u = 0$

$= \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \cos u \cdot du$

$= \frac{2}{\pi} [\sin u]_0^{\frac{\pi}{2}}$

$= \frac{2}{\pi} [\sin \frac{\pi}{2} - \sin 0] = \frac{2}{\pi} [1 - 0] = \frac{2}{\pi}$

61. $\int_0^1 \sqrt[3]{1+7x}\,dx$

Let $u = 1+7x$; $\frac{du}{dx} = 7$; $dx = \frac{1}{7}\,du$

When $x = 1$, $u = 1+7 = 8$

$x = 0$, $u = 1$

$= \int_1^8 \sqrt[3]{u} \cdot \frac{1}{7}\,du$

$= \frac{1}{7} \int_1^8 \sqrt[3]{u}\,du$

$= \frac{1}{7} \cdot \frac{3}{4} \cdot \left[u^{\frac{4}{3}}\right]_1^8$

$= \frac{3}{28}\left[8^{\frac{4}{3}} - 1\right] = \frac{3}{28}\left[16 - 1\right] = \frac{3}{28} \times 15$

$= 1.607$

65. $\int_1^2 \frac{e^{1/x}}{x^2}\,dx$

let $u = e^{\frac{1}{x}}$; $\frac{du}{dx} = e^{\frac{1}{x}} \cdot \left(-\frac{1}{x^2}\right)$; $dx = -\frac{x^2}{u} \cdot du$

when $x = 2$, $u = e^{\frac{1}{2}}$

$x = 1$, $u = e^{1}$

$= \int_{e^1}^{e^{\frac{1}{2}}} \frac{u}{x^2} \cdot \left(-\frac{x^2}{u}\right) du = -\int_{e^1}^{e^{\frac{1}{2}}} 1\,du = -[u]_{e^1}^{e^{\frac{1}{2}}}$

$= -\left[e^{\frac{1}{2}} - e^1\right]$

$= 1.0696$

67. $\int_{-\pi/4}^{\pi/4} (x^3 + x^4 \tan x)\, dx$

odd odd

$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} x^3\, dx + \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} x^4 \tan x\, dx$

$= 0$

let $f(x) = x^4 \tan x$

$f(x) = -f(-x)$

$\therefore f(x)$ is an odd function

$\therefore \int dx = 0$

69. $\int_0^{13} \frac{dx}{\sqrt[3]{(1+2x)^2}}$

let $u = (1+2x)^2$; $\frac{du}{dx} = 4(1+2x)$; $dx = \frac{du}{4(1+2x)}$

when $x = 13$, $u = (1+2\times13)^2 = 729$

$x = 0$, $u = (1+0) = 1$

$$\int_1^{729} \frac{1}{\sqrt[3]{u}} \cdot \frac{du}{4(1+2x)}$$

let $u = 1+2x$; $\frac{du}{dx} = 2$; $dx = \frac{du}{2}$

when $x = 13$, $u = 27$; $x = 0$, $u = 1$

$$\int_1^{27} \frac{1}{\sqrt[3]{u^2}} \cdot \frac{du}{2} = \frac{3}{2}\left[u^{\frac{1}{3}}\right]_1^{27} = 3$$

71. $\int_0^a x\sqrt{x^2 + a^2}\, dx \quad (a > 0)$

Let $u = x^2 + a^2$; $\frac{du}{dx} = 2x$; $dx = \frac{du}{2x}$

when $x = a$, $u = 2a^2$; $x = 0$, $u = a^2$

$$\int_{a^2}^{2a^2} x \cdot \sqrt{u} \cdot \frac{du}{2x}$$

$$= \frac{1}{2}\int_{a^2}^{2a^2} \sqrt{u}\, du = \frac{1}{2}\left[u^{\frac{3}{2}}\right]_{a^2}^{2a^2} = \frac{1}{3}\left[(2a^2)^{\frac{3}{2}} - (a^2)^{\frac{3}{2}}\right]$$

$$= 0.609\, a^3$$

73. $\int_1^2 x\sqrt{x-1}\,dx$

Let $u = x-1$, $\frac{du}{dx} = 1$; $dx = du$

$x = u+1$

when $x = 2$, $u = 1$

$x = 1$, $u = 0$

$$\int_0^1 (u+1)\sqrt{u}\,du = \int_0^1 u^{\frac{3}{2}} + u^{\frac{1}{2}}\,du$$

$$= \frac{2}{5}\left[u^{\frac{5}{2}}\right]_0^1 + \frac{2}{3}\left[u^{\frac{3}{2}}\right]_0^1$$

$$= \frac{16}{15}$$

77. $\int_0^1 \frac{e^z + 1}{e^z + z}\, dz$

Let $u = e^z + z$; $\frac{du}{dz} = e^z + 1$; $dz = \frac{du}{e^z + 1}$

when $z = 1$, $u = e + 1$

$z = 0$, $u = 1$

$$\int_1^{e+1} \frac{e^z + 1}{u} \cdot \frac{du}{e^z + z}$$

$$= \int_1^{e+1} \frac{1}{u}\, du$$

$$= [\ln|u|]_1^{e+1}$$

$$= \ln|e+1| = 1.3133$$